System Requirements

P02 : HerSafar

Student ID	Name
26100105	Maryam Rizwan
26100405	Kousar Pervaiz
26100352	Emaan Waleed
26100365	Muhammad
26100378	Muhammad Zain Naseer

Table of Contents

1. Introduction 2
2. System Actors 4
3. Functional Requirements 5
4. Non-functional Requirements / Quality Attributes 8
5. Security Requirements 9
6. Security Engineer 11
7. Use of Generative AI 13
8. Who Did What? 14
9. Review checklist 14

1. Introduction

Background and Motivation

Female domestic workers in Pakistan face major commuting challenges. They often travel long distances, with public transport being overcrowded, unreliable, and time-consuming, while private options are too costly, consuming much of their low wages. Men share some of these issues, but women also face added safety risks, making daily travel especially stressful.

Existing digital transport apps fail to address these needs. Most are English-intensive, text-heavy, and complex, requiring reading, typing, and map navigation. For women with low literacy and limited English, these apps are inaccessible, leaving them excluded from technology that could otherwise make commuting safer and more affordable.

Project Goal

The goal of this project is to design a **simple, voice-first commuting platform in Urdu and local languages** that ensures both secure and affordable travel for female domestic workers. By removing the need for reading or typing, the platform will make commuting support accessible to women with low literacy and minimal digital skills. The system will enable them to plan safe routes, arrange affordable carpools, and receive real-time spoken updates, while prioritizing secure travel through features like a buddy system, SOS emergency alerts, and trusted contact sharing.

Objectives

1. **Accessibility** – Design an inclusive, voice-based system in local languages that does not require reading, typing, or English proficiency.

2. **Safety** – Provide secure travel features such as a buddy system, trusted contact sharing, real-time updates, and an SOS emergency function.

3. **Affordability** – Offer cost-effective commuting options, including carpool coordination and optimized route planning.

4. **Ease of Use** – Build a simple, intuitive interface for users with minimal digital skills.

5. **Reliability** – Deliver real-time information and updates to ensure dependable commuting support.

Target Users

The **primary users** are female domestic workers in Pakistan, particularly those with low literacy and little to no English proficiency. **Secondary stakeholders** include their families, employers, and communities, who will benefit from the increased safety, reliability, and affordability of domestic workers' commutes.

Expected Impact

By introducing an easy-to-use, voice-based system, this project aims to empower female domestic workers by making their commutes safer, more affordable, and less stressful. It will reduce their vulnerability, improve their mobility, and support their participation in the workforce, ultimately contributing to their overall well-being.

2. System Actors

Actor Name	Description
Commuter	Commuters are the main users of the application, typically female domestic workers and other vulnerable groups. They use voice commands to find safe, affordable carpools, share their live location with trusted contacts, and access emergency safety features like SOS alerts.
Driver	Drivers are individuals offering rides along their daily routes. They interact with the system to register rides, receive optimized route suggestions, and collect secure payments through the app.
Trusted Contact	Trusted contacts are family members, friends, or guardians linked to a commuter's account. They receive live trip updates, alerts, and SOS notifications to ensure the commuter's safety during travel.
Admin	Admins manage the platform's overall operations. They verify user profiles, monitor system activity, respond to SOS alerts, handle dispute resolution, and ensure the platform remains safe and trustworthy.

3. Functional Requirements

Requirements of Commuter	
Sr#	**Requirement**
1	I want to use the app entirely through voice in Urdu or my regional language so that I can interact without needing to read or type.
2	I want to find safe and affordable carpool options so that I can save time and money on commuting.
3	I want to receive spoken updates about my trip (delays, route changes, driver arrival) so that I can stay informed without needing to read text.
4	I want to share my live trip details with trusted family members (buddy system) so that my family knows I am safe.
5	I want a one-tap SOS emergency button so that I can quickly get help in case of danger.
6	I want the app to work even with weak or no internet using SMS-based confirmations and SOS so that I am never stranded.
7	I want to make secure mobile payments so that I can contribute fairly to carpools without carrying cash.
8	I want to rate drivers/commuters and see ratings so that I can make safe ride choices.
9	I want to send and receive secure voice messages so that I can communicate even if I cannot type.

Requirements of Driver	
Sr#	**Requirement**
1	I want to offer rides for my daily route so that commuters can join and share costs.
2	I want to get AI-optimized routes so that I can save fuel and time while avoiding unsafe areas.
3	I want to get paid in the payment method chosen by the passenger (digital or cash), while making sure that every payment is securely recorded in my earnings log.

Requirements of Trusted Contacts of Commuter	
Sr#	**Requirement**
1	I want to receive live trip updates and location sharing so that I can track my family member's journey.
2	I want to be alerted instantly if SOS is triggered so that I can respond quickly or notify authorities.
3	I want to get notified if the commuter enters an unsafe area so that I can check in on their safety.

Requirements of Admin	
Sr#	**Requirement**
1	I want to verify user identities (CNIC/driving license checks) so that only trusted commuters and drivers use the system.

2	I want to add, update, or remove user accounts so that the platform remains safe and up to date.
3	I want to monitor digital and cash transactions so that payments are secure, transparent, and dispute-free.
4	I want to track rides in real time and receive anomaly alerts (unsafe detours, unusual behavior) so that I can ensure safety and intervene quickly when needed.
5	I want to monitor safety trends and update a database of unsafe areas so that the system can identify risks and warn users in real time.
6	I want to send community-wide safety alerts so that all commuters are informed about potential risks.
7	I want to analyze ride, payment, and user data through an analytics dashboard so that I can identify patterns, improve services, and make informed decisions.
8	I want to update fares, policies, and system rules so that the platform remains reliable and compliant with regulations.

4. Non-functional Requirements / Quality Attributes

Sr#	Requirements
1	The voice recognition system must achieve at least 85% accuracy for Urdu voice commands and 80% accuracy for regional dialects under normal noise conditions
2	The system must respond to voice commands within 6 seconds of voice input completion, even on 3G network connections
3	Battery consumption should not exceed 10% per hour of active usage on standard Android devices with a 3000mAh battery capacity.
4	The application must maintain 99.5% uptime availability, measured monthly, excluding planned maintenance windows not exceeding 4 hours per month.
5	The system should handle up to 100 active users simultaneously without any degradation in performance.
6	The app interface should be intuitive and simple enough for users with minimal digital literacy to understand without requiring more than 2 onboarding sessions.

5. Security Requirements

Sr #	Security Risks	Potential Losses	Controls
1	Insecure Authentication	Unauthorized access to user accounts, exposure of personal data	Use secure authentication, enforce strong passwords
2	Insufficient Input Validation	System crash, data leaks	Validate & Sanitize all inputs, implement input filters
3	Lack of Emergency Support	Physical harm	Implement an SOS feature: when triggered app will automatically sends live location + alert notification to the predefined emergency contacts
4	Improper Access Control	Data Manipulation	Role-based access control (RBAC)
5	Insecure Data Storage	Theft of sensitive information (e.g., location history, personal details), user privacy breaches, regulatory non-compliance	Encrypt sensitive data at rest and in transit, use secure databases, implement regular security audits and backups

6. Security Engineer

Name of the Security Engineer	Muhammad Zain Naseer

7. Use of Generative AI

Generative AI was used to refine language, structure content, and polish descriptions for clarity and professionalism. The project concepts and requirements were defined by the team; AI served only as a drafting and editing aid.

6. Who Did What?

Name of the Team Member	Tasks done
Emaan Waleed	1. Introduction + 3. Functional Requirements (small part) + 7. Use of Generative AI
Muhammad Zain Naseer	4. Non-functional Requirements / Quality Attributes
Maryam Rizwan	3. Functional Requirements
Muhammad	5. Security Requirements
Kousar Pervaiz	2. System Actors

7. Review Checklist

Section Title	Reviewer Name(s)
Introduction	Maryam Rizwan
Actors	Emaan Waleed
Functional Requirements	Muhammad Zain Naseer
Non-functional requirements	Kousar Pervaiz
Security Requirements	Emaan Waleed
Use of Generative AI	Kousar Pervaiz